



Health Science Department
Masters in Applied Health Informatics

Course Title: HHA 503: Regulations, Confidentiality, Privacy and Security (3 credits)

Course Time: (Asynchronous):

Instructor: Blake Benz
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Office Hours: Available upon request

Description: This course provides foundational knowledge in the laws, regulations, policies and procedures related to confidentiality, privacy, and security on all levels of health-related information and infrastructures. Special emphasis on interoperability, HIPAA/HITECH Privacy Rule and Security Standards, meaningful use, and IT security forensics.

Goal: Provide a working knowledge (conceptual framework and the practical applications) of security and privacy regulations and standards as required of health informatics professionals.

Learning Objectives: Upon completion of this course, students will be able to:

- 1) Describe the original intention of the Health Insurance Portability and Accountability Act (HIPAA) as it relates to ensuring the privacy of Protected Patient Information (PHI) and securing this data from the perspective of Covered Entities (CEs) and Business Associates (BAs) and the enforcement of these rules.
- 2) Analyze the acceptable uses of PHI and how these uses are protected by the HIPAA Privacy Rule of HIPAA which specifies patient rights.
- 3) Discuss methods to secure PHI including administrative, technical and physical safeguards.
- 4) Outline standards created by HIPAA for better, more efficient communication within the health care industry. These standards include transaction / code sets and unique identifiers.
- 5) Understand the Health Information Technology for Economic and Clinical Health (HITECH), its further effect on protecting the privacy and security of PHI, especially as it relates to electronic records.
- 6) Evaluate the effectiveness of Meaningful Use and Interoperability on the adoption and efficacy of Electronic Health Records (EHRs)

7) Critique cybersecurity strategies and how these approaches influence the privacy and security of PHI.

8) Analyze the effectiveness of recent changes and additions of privacy and security regulations.

9) Research current issues in healthcare privacy and security including Artificial Intelligence (AI), blockchain technology, big data and telemedicine.

Students are required to prepare for class by doing the assigned reading, viewing short video lectures, and completing assignments.

Below are links to online resources offering data, research and information on healthcare privacy and security in the US. Feel free to sign up to receive periodic emails/newsletters from some or all of these resources. This is not an exhaustive list – healthcare privacy and security is a vast topic, and there will be other resources you discover on your own that you find useful as well; the list below is a starting point.

- <https://pubmed.ncbi.nlm.nih.gov/>
- <https://scholar.google.com>
- <https://journal.ahima.org/privacy-and-security>
- <https://www.hipaajournal.com/>
- <https://www.healthcareitnews.com/category/resource-topic/privacy-security>
- <https://healthitsecurity.com/>

Required Textbook: Herold R, Beaver K. The Practical Guide to HIPAA Privacy and Security Compliance. Second edition. Auerbach; 2021.

Grading Scale:

A (100-93) B (86-83) C (76-73) D (66-60)
A- (92-90) B- (82-80) C- (72-70) F (<60)
B+ (89-87) C+ (79-77) D+ (69-67)

Evaluation: *Information and grading rubrics detailing the group and individual projects and presentations will be posted on Brightspace*

2 Discussion Board Posts (10% each)	20%
Individual Video Presentation (20% for submission, 10% for comments on 2 presentations)	30%
Group Project/Presentation	20%
Term Paper	<u>30%</u>
Total	100%

Participation in Discussion Board on Brightspace (20%)

Discussion prompts will be posted **two times** during the semester. Discussion posts and responses are due as per the calendar at the end of the syllabus. The maximum length for all posts is **300 words**, although there is not a minimum length. I require a minimum of **two references** for the initial post. You must **respond to at least two other students** and must include at least **one reference in the responses**. I expect all students to share their opinions, discuss, and even sometimes debate the questions or topics posted. Strong discussion board threads include varying opinions, and as long as everyone remains respectful, I look forward to robust, thought-provoking discussions. It is OK to respectfully disagree with each other (or with me!) I emphasize quality over quantity, and in fact being efficient in your communications is an important skill in the real world. On the other hand, I am looking for value in your posts, and it is very clear when a post does not really say anything of value. Also, if you state facts, be prepared to defend their validity!

Student Video Presentation and Responses (30%)

The purpose of this assignment is to share a current/relevant healthcare privacy/security article or news story and summarize it for the class. Each student will present a short report to the class on the student-selected article found in a reputable journal, newspaper, or magazine. You may use a site from the Resources section above or check with me to make sure that your source is acceptable. **NOTE: Please enter the article you have chosen in the Excel file in Brightspace in the Content section. Duplicate selections will not be allowed; coordinate with your classmates as appropriate.**

You can also bring in additional literature and/or articles from your own research relevant to the topic. The more research you do, the more you will learn, and the more points you will earn!

Your presentation should be between 6 and 8 minutes long. Make sure to discuss what struck you about the topic and/or your reaction/perspective on the issues/events in the article. You will be expected to synthesize all material thoroughly, and provide your analysis of your reading (i.e., discuss the key theme(s), message(s) or finding(s) in the paper/article and how it is relevant to the topic of discussion).

Bear in mind that the point of this exercise is to promote critical thinking skills such as interpretation, analysis, and discussion of the content. Feel free to prepare a slide/presentation to summarize the article's content and do this interpretation and analysis.

Please upload your completed video to the discussion board topic "Individual Video Presentation". The thread title should be your first and last name. For example, my thread title would be "Blake Benz". **Also, your video must be less than 2 MB.** This will make it easier for the classmates assigned to give feedback to find your video.

You will then be assigned to watch the videos for two other classmates and give feedback on the presentations. Your assigned classmates are in the content section of Brightspace with the file name "Assigned Individual Presentations to Watch". Your feedback should be limited to 300 words and should include one strength of the presentation and one area in which the

presentation could be improved. Please give details on why the strength is a strength and the how the area for improvement can be improved. Points will be awarded based on details given. Please be respectful! I am looking for constructive criticism and thoughtful analysis.

Group Project/Presentation (20%)

Based on the information you have learned about the first 2 HIPAA titles, create a scenario in which a HIPAA violation may have occurred. Each group may pick from one of the following basic scenarios. You must create a detailed scenario which may be presented by one person. The description of the scenario should last from 3-5 minutes. Then, the rest of your group must role play the “perpetrator,” the privacy officer, witnesses, patients, patient family members, and other employees. You do not need to have all the above persons as part of the role play except the perpetrator and the privacy officer, but one student may play more than one part of the role play. The privacy officer “interview” should include who the person is, how they may have been involved in the incident and any other pertinent information. Each interview should last at least 1 minute, and the total “investigation” should last from 10 to 12 minutes. At the end of the role play, the privacy officer must determine if the perpetrator is guilty of a HIPAA violation and determine the penalty. This final summary should last 2-3 minutes. The entire role play should last between 15 and 20 minutes.

Two percent of your final overall grade will be based on the peer evaluation form which can be found on the Assignment Materials and Examples tab in Brightspace. This is to ensure that all students participate fully in this assignment.

Possible scenarios (first-come, first-served bases via sign-up in Brightspace for scenario selection for each group)

- 1) The director of a department knows his mother is in a hospital in the same system as the hospital he works at. He has been talking to some of his co-workers about how it seems his mother’s health has improved.
- 2). The CEO wants to have a picture taken of the ER staff during hospital week. She asks an employee who is known to be a good photographer to take the picture in the emergency department.
- 3) A physician is discussing a patient’s status with a group of physicians in a crowded cafeteria at the hospital. There are many other people who may or may not have overheard the conversations.
- 4) A security staff is approached by a person who says he is there to visit his wife. After a few minutes of conversation, the man leaves. It is later discovered the man verbally assaulted his wife in her hospital room.
- 5) A patient calls the hospital stating they received the discharge paperwork for another patient. It is uncertain how they got it and if the other patient got the correct paperwork.

6) A nurse is speaking to another nurse about a particular patient's care plan. It is later learned that the patient's boyfriend found out the patient had a diagnosis the patient did not want him to know about.

Term Paper (30%)

Sign up next to a topic in the Excel file "Term Paper" on Brightspace. **You will write a paper not longer than 4 pages (excluding title page and references). The paper should be double-spaced, 12 font, and Times New Roman.** Please include **AT LEAST 5 references**. The more articles you find, the more you will learn and the better you will do! Find articles, research, and other types of information on the topic you choose and thoroughly consider best practices, avoiding breaches, and other nuances on the topic that you can describe, research and analyze. Search the Stony Brook University Library, Google Scholar and PubMed to find articles. Find something novel and focus on that specific topic. Don't assume, for example, that if you choose to write on the "technical, physical and/or administrative safeguards" topic that you must define all the safeguards in detail. Instead, briefly define all points at a high level, then choose one part of the subject to focus more in depth on and find some examples on the topic and analyze these examples. At max, one page should be a description of your topic. The rest should be analysis. Find an angle we have not discussed during this term and your paper will be even stronger! Only write your name next to one of the topics listed **UNLESS** you get a new topic approved by me. **If you write about a topic not on the list and I do not approve it, you will earn a zero!**

Course Outline:

Week	Required Readings	Topics / Assignments
Week 1: 7/6 - 7/12	Herold & Beaver, Chapters 1, and 3-6 Watch HIPAA Law Summary Video: https://study.com/academy/lesson/video/what-is-hipaa-definition-requirements-laws.html Review PowerPoint slides: "HIPAA title 1 and Title II and HIPAA Title I, cont." AND "Title II – Security, Transaction and Code Sets"	Original HIPAA Law <i>Optional Zoom Meeting on 7/6 at 7p</i> Discussion Board 1 due on 7/9 at 11:55p with 2 responses due on 7/12
Week 2: 7/13 - 7/19	Herold & Beaver, Chapters 2, 7-9, 17, and 18 Watch HITECH Act Video: https://www.youtube.com/watch?v=OM8tmPy-ApI	HITECH and Meaningful Use (2009) Discussion Board 2 due on 7/16 at 11:55p with 2 responses due on 7/19 at 11:55p

	Review PowerPoint slides: “Title II – Unique Identifier and Enforcement Rules, HITECH Intro AND HITECH, cont.” AND “HITECH Meaningful Use”	
Week 3: 7/20 - 7/26	Herold & Beaver, Chapters 10-11, and 19-21 Review PowerPoint slide “Effects of HITECH” Adler-Milstein J, Jha AK. HITECH Act Drove Large Gains In Hospital Electronic Health Record Adoption. <i>Health Affairs</i>. 2017;36(8):1416-1422 Kim H, Lee J. The impact of health IT on hospital productivity after the enactment of HITECH Act. <i>Applied economics letters</i>. 2020;27(9):719-724. doi:10.1080/13504851.2019.1644433 Lin YK, Lin M, Chen H. Do Electronic Health Records Affect Quality of Care? Evidence from the HITECH Act. <i>Information systems research</i>. 2019;30(1):306-318	Effects of HIPAA and HITECH <i>Optional Zoom Meeting on 7/20 at 7p</i> Student Video topic due on 7/22 @ 11:55p Group Presentation AND Peer Evaluation due on 7/26 @ 11:55p
Week 4: 7/27 - 8/2	Herold & Beaver, Chapters 12-16 Listen to Cybersecurity podcast: https://www.youtube.com/watch?v=3cjFA76pFAo Review PowerPoint slide “Other Security Topics” Coventry L, Branley D. Cybersecurity in healthcare: A narrative review of trends, threats and ways forward. <i>Maturitas</i>. 2018 Jul;113:48-52. doi: 10.1016/j.maturitas.2018.04.008. Epub 2018 Apr 22. PMID: 29903648. Tully J, Selzer J, Phillips JP, O'Connor P, Dameff C. Healthcare Challenges in the Era of Cybersecurity. <i>Health Secur</i>. 2020	Cybersecurity in Healthcare Sign up for Term Paper due on 7/31 at 11:55p Individual Video Presentation due on 8/2 at 11:55p

	May/Jun;18(3):228-231. doi: 10.1089/hs.2019.0123. PMID: 32559153.	
Week 5: 8/3 - 8/9	Herold & Beaver, Chapters 25, 26 Watch Why Healthcare Privacy Is Facing a Perfect Storm: 3 Trends You Need to Know Video: https://www.youtube.com/watch?v=Xzlyhe2r5bQ Review PowerPoint slide “Recent Developments” YARAGHI N, GOPAL RD. The Role of HIPAA Omnibus Rules in Reducing the Frequency of Medical Data Breaches: Insights From an Empirical Study. The Milbank quarterly. 2018;96(1):144-166. doi:10.1111/1468-0009.12314	<i>Optional Zoom Meeting on 8/3 at 7p</i> Changes in privacy, security and interoperability since 2009 (Omnibus rules) Response to two Individual Video Presentations due on 8/9 at 11:55p
Week 6: 8/10 - 8/15	Price WN 2nd, Cohen IG. Privacy in the age of medical big data. Nat Med. 2019 Jan;25(1):37-43. doi: 10.1038/s41591-018-0272-7. Epub 2019 Jan 7. PMID: 30617331; PMCID: PMC6376961. Hamm JM, Greene C, Sweeney M, et al. Telemedicine in the emergency department in the era of COVID-19: front-line experiences from 2 institutions. <i>Journal of the American College of Emergency Physicians Open</i>. 2020;1(6):1630-1636 Wang L, Liu X, Shao W, et al. A Blockchain-Based Privacy-Preserving Healthcare Data Sharing Scheme for Incremental Updates. <i>Symmetry (Basel)</i>. 2024;16(1):89- Khalid N, Qayyum A, Bilal M, Al-Fuqaha A, Qadir J. Privacy-preserving artificial intelligence in healthcare: Techniques and applications. Comput Biol Med. 2023 May;158:106848. doi: 10.1016/j.combiomed.2023.106848. Epub 2023 Apr 5. PMID: 37044052.	Special Privacy and Security Topics, Part I Term paper due on 8/13 at 11:55p

	Morley J, Machado CCV, Burr C, Cows J, Joshi I, Taddeo M, Floridi L. The ethics of AI in health care: A mapping review. Soc Sci Med. 2020 Sep;260:113172. doi: 10.1016/j.socscimed.2020.113172. Epub 2020 Jul 15. PMID: 32702587.	
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Note to students: This course has been developed for the serious student of healthcare informatics. Information will be presented in sequential progression for the progressive development of requisite knowledge, skills, and competencies. This requires sustained and timely student preparation and participation. The professor will provide feedback and monitor progress; students are expected to be prepared to discuss the materials assigned for each class whether face-to-face or on-line.

Brightspace: Students will find all course documents or announcements on Brightspace. You should check Brightspace several times a week for updates or changes to class and/or assignments. Please check student personal information to be sure that the email address is current. For problems with logging in, go to the helpdesk in the Main Library SINC Site or the Union SINC Site. You may also call: 631-632-9602 or email: helpme@ic.sunysb.edu

Plagiarism Policy: Plagiarism is simply the use of others' words and/or ideas without clearly acknowledging their source. As students, you are learning about other people's ideas in your course texts, your instructors' lectures, in-class discussions, and when doing your own research. When you incorporate those words and ideas into your own work, it is of the utmost importance that you give credit where it is due. Plagiarism, intentional or unintentional, is considered academic dishonesty and all instances will be reported to the Academic Judiciary. To avoid plagiarism, you must give the original author credit whenever you use another person's ideas, opinions, drawings, or theories as well as any facts or any other pieces of information that are not common knowledge. Additionally, quotations of another person's actual spoken, or written words; or a close paraphrasing of another person's spoken or written words must also be referenced. Accurately citing all sources and putting direct quotations – of even a few key words – in quotation marks are required. For further information on plagiarism and the policies regarding academic dishonesty go to the Academic Judiciary website at <http://naples.cc.sunysb.edu/CAS/ajc.nsf>.

Generative Artificial Intelligence built on a Large Language Model like ChatGPT is allowed however, its use MUST be cited in your references. Also, please do not use these technologies to write 100% of your assignments. Please put your writing through an LLM before submitting and, if necessary, rewrite some of the assignments in your own words.

Student Accessibility Support Center Statement:

If you have a physical, psychological, medical or learning disability that may impact your course work, please contact Student Accessibility Support Center, ECC (Educational Communications Center) Building, Room 128, (631)632-6748. They will determine with you what accommodations, if any, are necessary and appropriate. All information and documentation is confidential.

Students who require assistance during emergency evacuation are encouraged to discuss their needs with their professors and Student Accessibility Support Center. For procedures and information go to the following website: <http://www.stonybrook.edu/ehs/fire/disabilities>.

Academic Integrity Statement:

Each student must pursue his or her academic goals honestly and be personally accountable for all submitted work. Representing another person's work as your own is always wrong. Faculty is required to report any suspected instances of academic dishonesty to the Academic Judiciary. Faculty in the Health Sciences Center (School of Health Technology & Management, Nursing, Social Welfare, Dental Medicine) and School of Medicine are required to follow their school-specific procedures. For more comprehensive information on academic integrity, including categories of academic dishonesty please refer to the academic judiciary website at http://www.stonybrook.edu/commcms/academic_integrity/index.html

Critical Incident Management:

Stony Brook University expects students to respect the rights, privileges, and property of other people. Faculty are required to report to the Office of University Community Standards any disruptive behavior that interrupts their ability to teach, compromises the safety of the learning environment, or inhibits students' ability to learn. Faculty in the HSC Schools and the School of Medicine are required to follow their school-specific procedures. Further information about most academic matters can be found in the Undergraduate Bulletin, the Undergraduate Class Schedule, and the Faculty-Employee Handbook.